Bessel's Equation and Bessel Function

The most important differential equation

$$x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(x^2-n^2)y=0$$

where n is a constant (not necessarily an integer) is known in the literature as **Bessel's Diiferential Equation of order n**. It is a linear ordinary differential equation of 2nd order. This equation has a wide range of applications such as heat transfer, vibration, stress analysis and fluid mechanics.

- The solution of Bessel's equation is called Bessel function. Bessel functions are of two kinds (first and second kind). At first, they were defined by the mathematician Daniel Bernoulli and then generalized by Friedrich Bessel. And the functions are associated with a wide range of problems in important areas of mathematical physics especially in wave propagation and signal processing.

There are various methods to solve Bessel Differential equation. And two of them are popular:

1. Laplace transform
2. Power Series method

- **Bessel Function of the first kind** of order **n** is denoted by $J_n(x)$ and defined as,

$$J_n(x)=\sum_{r=0}^{\infty}\frac{(-1)^r}{r!\ \Gamma(n+r+1)}\left(\frac{x}{2}\right)^{n+2r}$$

- **Bessel Function of the first kind** of order **-n** is denoted by $J_{-n}(x)$ and defined as,

$$J_{-n}(x)=\sum_{r=0}^{\infty}\frac{(-1)^r}{r!\ \Gamma(-n+r+1)}\left(\frac{x}{2}\right)^{-n+2r}$$

Hence the complete solution of the Bessel's equation can also be expressed in the form

$$y=AJ_n(x)+BJ_{-n}(x)$$

and $J_{-n}(x)=(-1)^nJ_n(x)$

- Generating function of Bessel function, $J_n(x)$:

$$e^{\frac{1}{2}x\left(t-\frac{1}{t}\right)}=\sum_{n=-\infty}^{\infty}t^n\ J_n(x)$$

Recurrence formulae for Bessel function, $J_n(x)$:

i) $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$

ii) $\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$

iii) $J_n(x) = \frac{x}{2n}[J_{n-1}(x) + J_{n+1}(x)]$

iv) $J_n'(x) = \frac{1}{2}[J_{n-1}(x) - J_{n+1}(x)]$

v) $xJ_n'(x) = nJ_n(x) - xJ_{n+1}(x)$

vi) $xJ_n'(x) = -nJ_n(x) + xJ_{n-1}(x)$

<u>Example-1</u>: Use $xJ_n'(x) = nJ_n(x) - xJ_{n+1}(x)$ to prove that,

(i) $J_0'(x) = -J_1(x)$

(ii) $J_2(x) = J_0''(x) - \frac{J_0'(x)}{x}$

<u>Example-2</u>: Use $J_n'(x) = \frac{1}{2}[J_{n-1}(x) - J_{n+1}(x)]$ to prove that $2J_0''(x) = J_2(x) - J_0(x)$

<u>Example-3</u>: Use $J_n(x) = \frac{x}{2n}[J_{n-1}(x) + J_{n+1}(x)]$ to prove that $J_3(x) = \left(\frac{8-x^2}{x^2}\right)J_1(x) - \frac{4}{x}J_0(x)$

<u>Example-4</u>: Use $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$ to show that $\int_0^x x^{n+1} J_n(x)dx = x^{n+1} J_{n+1}(x)$

<u>Example-5</u>: Use $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$ to prove that,

(i) $\frac{d}{dx}[xJ_1(x)] = xJ_0(x)$

(ii) $\int_0^b xJ_0(ax)dx = \frac{b}{a}J_1(ab)$

<u>Example-6</u>: Prove the following

(i) $J_{\frac{1}{2}}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\sin x$

(ii) $J_{-\frac{1}{2}}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\cos x$

(iii) $J_{\frac{3}{2}}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\left(\frac{\sin x}{x} - \cos x\right)$

(iv) $J_{\frac{-3}{2}}(x) = -\sqrt{\left(\frac{2}{\pi x}\right)}\left(\frac{\cos x}{x} + \sin x\right)$

Example-7: Use $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$ to show that

$$\int x^4 J_1(x)dx = x^4 J_2(x) - 2x^3 J_3(x) + c$$

Hermite's Equation and Hermite Polynomial

The differential equation $\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2ny = 0$

where n is a constant, is called **Hermite's Diiferential Equation**.

Hermite polynomials are the solutions of Hermite's differential equation.

Charles Hermite (1822-1901) gave us these polynomials, famous in the quantum mechanics of the harmonic oscillator. These functions can be used to specify basis states in quantum mechanics.

Hermite polynomial of order **n** is defined by

$$H_n(x) = \sum_{r=0}^{N} (-1)^r \frac{n!}{r!\,(n-2r)!}(2x)^{n-2r}$$

where $N = \begin{cases} \frac{n}{2}, & n = even \\ \frac{n-1}{2}, & n = odd \end{cases}$

Hermite Polynomials:

$$\begin{aligned}
H_0(x) &= 1, \\
H_1(x) &= 2x, \\
H_2(x) &= 4x^2 - 2, \\
H_3(x) &= 8x^3 - 12x, \\
H_4(x) &= 16x^4 - 48x^2 + 12, \\
H_5(x) &= 32x^5 - 160x^3 + 120x, \\
H_6(x) &= 64x^6 - 480x^4 + 720x^2 - 120, \\
H_7(x) &= 128x^7 - 1344x^5 + 3360x^3 - 1680x, \\
H_8(x) &= 256x^8 - 3584x^6 + 13440x^4 - 13440x^2 + 1680, \\
H_9(x) &= 512x^9 - 9216x^7 + 48384x^5 - 80640x^3 + 30240x, \\
H_{10}(x) &= 1024x^{10} - 23040x^8 + 161280x^6 - 403200x^4 + 302400x^2 - 30240.
\end{aligned}$$

- Generating function of Hermite Polynomial, $H_n(x)$:

$$e^{2tx-t^2} = \sum_{n=0}^{\infty} \frac{t^n}{n!} H_n(x)$$

Recurrence formulae for Hermite Polynomial, $H_n(x)$:

i) $H_n'(x) = 2nH_{n-1}(x),\ n \geq 1$ and $H_0'(x) = 0$

ii) $H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x),\ n \geq 1$ and $H_1(x) = 2xH_0(x)$

iii) $H_n'(x) = 2xH_n(x) - H_{n+1}(x)$

iv) $H_n''(x) - 2xH_n'(x) + 2nH_n(x) = 0$

<u>Example 1</u>: Express $f(x) = \frac{x^4}{8} - 3x + 2$ in terms of Hermite Polynomials.

<u>Example 2</u>: Prove that $x^3 = \frac{1}{8}[H_3(x) + 6H_1(x)]$.

<u>Example 3</u>:

(a) Use the generating function for Hermite polynomials to prove that

(i) $H_{2n}(0) = (-1)^n \frac{(2n)!}{n!}$

(ii) $H_{2n+1}(0) = 0$

(b) Use the result obtained from 3(a) to evaluate the following.

(i) $H_{10}(0)$

(ii) $H_{101}(0)$

<u>Example 4</u>:

(i) Use the generating function for Hermite polynomials to prove that

if $m < n$, then $\frac{d^m}{dx^m}[H_n(x)] = \frac{2^m n!}{(n-m)!} H_{n-m}(x)$

(ii) Evaluate $\frac{d^6}{dx^6}[H_8(x)]$ by using the result of 4(i) as formula and then prove that the same result will be obtained by using calculus.